CO2-AT1

1) a) Input Conditions;

- Course registration : Registered or not registered
- Attendance : 0% - 100%, with a minimum of 75%.
- Examination fee : Paid or unpaid
- Register number : Exactly 12 digits.

Equivalence classes :

Input	Valid Class	Invalid class
Course	Registered	Not registered
Attendance	75% - 100%	Below 75%, below 0% or above 100%
Fee	Paid	unpaid
Registered number	12 digits	Fewer / more than 12 digits

Expected output : Registration is accepted only when all conditions are satisfied. Otherwise, the system displays the appropriate rejection, Payment request or error message

b) Equivalence class Partitioning (ECP):

Input are diuided into valid and invalid group and one representative value is selected from each group.

Boundary Value Analysis (BVA):

Values near the 75% attendance limit are tested.

Test Case	Attendance	Expected result
TC1	74%.	Registration rejected
TC2	75%.	Registration accepted if all other conditions are valid
TC3	76%.	Registration accepted if all other conditions are valid.

c)

Course registered	Attendance ≥ 75%	Fee Paid	Register number Valid	Expected result
Yes	Yes	Yes	Yes	Accepted
No	Yes	Yes	Yes	Rejected
Yes	No	Yes	Yes	Rejected
Yes	Yes	No	Yes	Payment required
Yes	Yes	Yes	No	Error displayed

d)

Test	Attendance	Register Number	Expected result
TC1	82%	123456789012	Accepted if other conditions are valid
TC2	40%	123456789012	Rejected
TC3	105%	123456789012	Invalid attendance
TC4	90%	12345678912	Invalid register number

Test : Enter ,
Ifsc - number

on : Choose Valid ,
undary Values

ion : Compare actual
expected outputs

cation : Ensure all
and important
are tested

ogged Out
d credentials and OTP)

Logged In
beneficiary and enter
id amount)
in

Network fail
↓
Transfer failed

Valid transitions : Logged out -> logged in -> Transfer in Progress -> Transfer Successful / transfer fail.

Invalid transitions

Transferring money without login, transferring without Selecting a beneficiary, and accessing a locked account without authorized recovery.

Test cases:

Test	Action	Expected result
TC1	Enter Valid Credenticals and OTP	Logged In
TC2	Enter incorrect OTP three times	Account Locked
TC3	Select beneficiary and Valid amount	Transfer in Progress
TC4	Disconnect network during transfer	Transfer failed.

b)

Test	OTP attempts	Expected result
TC1	Correct OTP on first attempt	Login Successful
TC2	Two incorrect OTPs, then correct OTP.	Login Successful
TC3	Three incorrect OTPs	Account locked.

c) Factors - Device (Laptop / Mobile), Browser, Network

Test	Device	Browser	Network
TC1	Laptop	Chrome	Wi-Fi
TC2	Laptop	Edge	Mobile Data
TC3	Mobile	Chrome	Mobile Data
TC4	Mobile	Edge	Wi-Fi

d) Use-case testing : Verifies complete user activities, while exploratory testing allows tester to explore the application to find unexpected defects.

1. Duplicate transactions :
2. Session expiry
3. Network interruption
4. Incorrect beneficiary

e) COTS means commercial off-the-Shelf, a ready made Software component.

1. Integration testing
2. Compatibility testing
3. Reliability testing
4. Security testing.

3)a) The Program has three decision nodes :

- D1 : amount <= 0
- D2 : amount > balance
- D3 : amount > dailylimit .

Control flow :

Start -> Read inputs

↓

D1 : amount ≤ 0 ?

Yes

↓

Invalid amount

No

↓

D2 : amount > balance?

Yes

Insufficient Balance

No

D3 : amount > dailylimit ?

(All outcomes lead to ⊘ top)

b)

Test	Password input	Expected result
TC1	Correct on first attempt	Login Successful
TC2	Wrong, then correct	Login Successful
TC3	Wrong twice, then correct	Login Successful
TC4	Three incorrect Passwords	Account locked
TC5	No Password attempt	No login; no incorrect lock message.

c) **Defect**: If the correct Password is entered on the third attempt, the Program may display both "Login Successful" and "Account locked"

Correction: Use a Boolean Variable to track Successful login.

Regression test cases :

- Correct Password on the first attempt —> Login Successful
- Correct Password on the third attempt —> Login Successful only
- Three incorrect Passwords —> Account locked only

d) • Condition coverage : Test both true and false outcomes of count < 3 and the Password comparison.

- Branch Coverage : Execute both outcomes of each decision.
- Path testing : Test Successful login on the first, Second and third attempts, and failure on all attempts
- Boundary - Value testing : Check the first, Second and third attempts, especially Success on the third attempt.

e) Possible mutations :

- Change count < 3 to count <= 3.
  A test should detect an extra attempt
- Change Password == Correct Password to Password != Correct Password.
  Login tests should fail.
- Remove BREAK. The Program may Continue

Test adequacy : A good test suite should detect these mutations and cover all important branches and boundary cases.

4)a) Structure :

- Loop Structure : WHILE count < 3
- Decision structures : IF Password == Correct
  IF count == 3.

Control flow :

Start ; Count = 0

↓

Count < 3 ?

Yes

↓

Read Password

↓

Password correct ?

↓

- Yes -> login
- NO -> repeat

NO

↓

Check count = = 3

(True -> Account locked)

b)

Test	Password input	Expected result
TC1	Correct on first attempt	Login Successful
TC 2	Wrong, then correct	Login Successful
TC 3	Wrong twice, then Correct	Login Successful
TC 4	Three incorrect Passwords	Account locked
TC5	No Password attempt	No login

c) **Defect** : If the correct Password is entered on the third attempt, the Program may display both "Login Successful" and "Account locked

Correction : Use a Boolean Variable to track Successful login.

Regression test cases :

=> Correct Password on the first attempt
—> Login Successful

=> Correct Password on the third attempt
—> Login Successful only

=> Three incorrect Passwords —> Account locked only

d)

- **Condition Coverage** : Test both True and False outcomes of Count < 3 and the Password comparison
- **Branch Coverage** : Execute both outcomes of each decision
- **Path testing** : Test Successful login on the first, Second and third attempts, and failure on all attempts.
- **Boundary - Value testing** : Check the first, Second and third attempts, especially Success on the third attempt

e) **Possible mutations** :

- Change Count < 3 to Count <= 3. Test fail
- Change Password == Correct Password to Password != Correct Password. Test fail
- Remove BREAK. Program may Continue
- Remove the login Success check. The System may lock.

5) a) Control flow :

Flow : Start -> Read N -> Outer loop -> Initialize total = 0 -> Read M -> Inner loop -> Read mark -> Add mark to total -> Print total -> Repeat -> Stop

- Outer loop : Processes each student
- Inner loop : Process each Subject
- Initialization : total = 0
- Definition and uses : Variables are assigned values and used in loop conditions or total calculation

b)

Test	Input	Expected result
1	N = 0	No results
2	N = 1, M = 1, mark = 80	Total = 80
3	N = 2, each has marks 70,80	Totals = 150, 150
4	N = 50, M = 10	All results Calculated
5	N = 51	Invalid input
6	M = 11	Invalid input

c)

Variable	Definition and use
N	Read and used in outer loop
M	Read and used in inner loop
i, j	Initialized, incremented and checked
mark	Read and added to total
total	Reset, update, Printed

d) ECP : Valid & Students : 1 - 50; Subjects : 1-10; marks 0-100;

BVA : Test Students 0, 1, 50; Subjects 0, 1, 10, 11;

marks - 1, 0, 100, 101

Error guessing : Test negative marks, marks above 100, Zero Subjects and incorrect total resetting

e) 1. **Statement coverage** : Execute every Statement.

2. **Branch coverage** : Test true and false loop conditions

3. **Loop coverage** : Test zero, one, typical and maximum iteration

4. **Data flow Coverage** : Check variable initialization, updates and uses.